

Wireless Programmable Logic Calculator

Example: Program the circuit to compute $X = A'BC + BC'D + AD + B'$

For $A=0, B=1, C=1, D=0 \Rightarrow X=1 \Rightarrow$ Output LED lights up

Built: Designed and built a four input logic calculator that can be programmed wirelessly from a PC

Solved: Resolved a communication issue between the Bluetooth module and the microcontroller by writing custom firmware.

Result: Enabled wireless programming of the circuit's logic function without rewiring

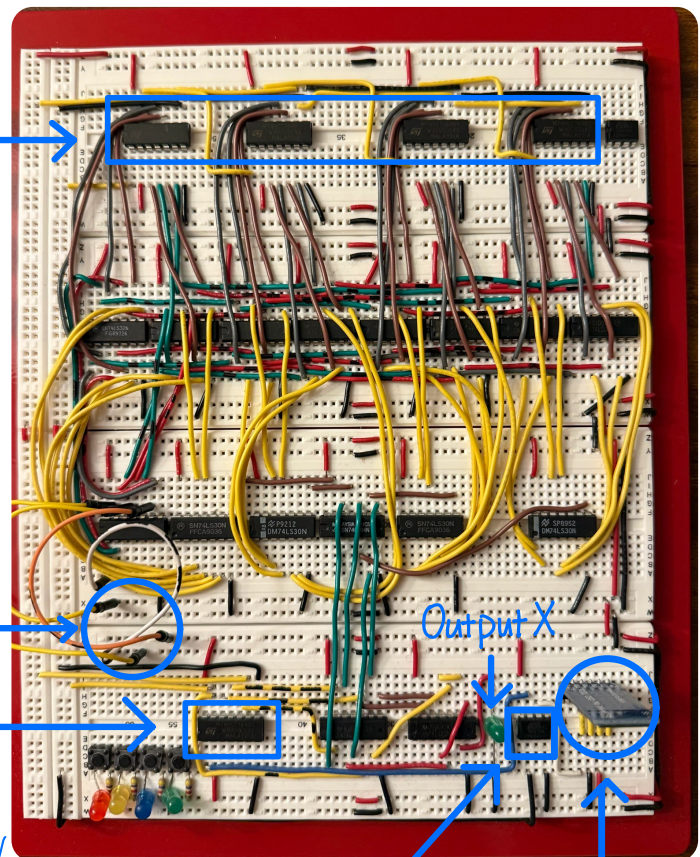
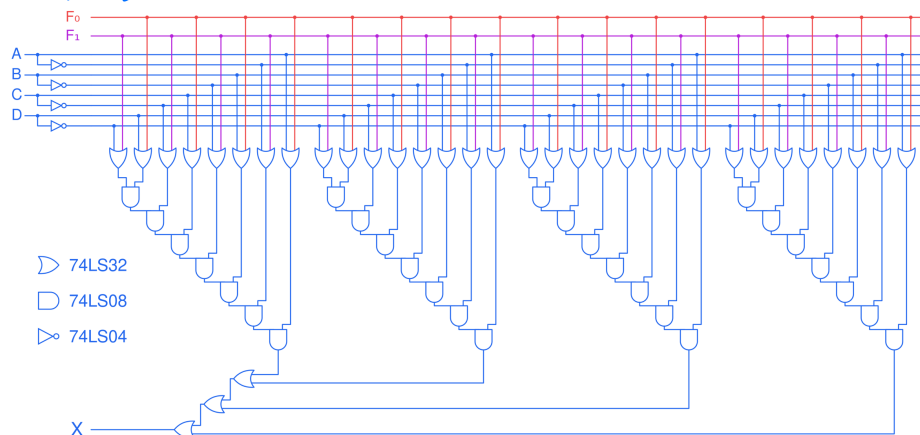
Shift registers

The shift registers store and output binary configuration bits to the logic gates, selecting variables to implement the desired formula

Inputs A-D (0/1)

Shift register

The area outside the boxes is the programmable logic calculator Diagram below



Bluetooth

Microcontroller

Converts bluetooth data into clock and data signals for shift registers